



Requirements

BIS601

Analysis Phase

- Expanding vision described in system request into thorough, detailed understanding of exactly what new system needs to do
- Output of planning phase (system request, feasibility study, project plan – business goals, project scope, work plan) are key inputs into analysis phase
- Work extensively with business users of new system to understand their needs

Analysis Phase (cont.)

- Basic process:
 - Understand existing situation (as-is system)
 - Identify improvements
 - Define requirements for new system (to-be system)

Analysis Phase (cont.)

- Critical thinking skills
 - To understand issues and develop new and improved business processes that are supported by information technologies
 - Essential in examining results of requirements discovery and translating those requirements into concept for new system

Analysis Phase (cont.)

- Example
 - User: The new system should “eliminate inventory stock-outs”
 - Circumstances leading to stock-outs (e.g., supplier order not placed in timely ways)
 - Actual business practices that lead to these circumstances (e.g., on-hand inventory levels are updated only once a week, delays occur in identifying best supply source, delays occur in receiving approval of supply order)

Analysis Phase (cont.)

- Requirements based on issues truly needed to be fixed
 - The system will update on-hand inventory levels twice per day
 - The system will produce out-of-stock notification immediately when item quantity on hand reaches item reorder point
 - The system will include recommended supplier with every out-of-stock notification

Analysis Phase (cont.)

- The system will produce supply purchase order that is sent to appropriate manager for approval
- The system will send approved purchase order to supplier via secure electronic communication

Analysis Phase (cont.)

- Unrealistic to expect true ready-made requirements during a few conversations with business users
- Analyst digs into the situation and discover requirements
- Final deliverable: system proposal – document compiling detailed requirements definition statement, use cases, process models, data models, revised feasibility analysis, and work plan
- Review project's business value

Analysis Phase (cont.)

- During requirements determination, to-be system concept is easy to change because little work has been done
- Iterative approaches: small batches of requirements can be identified and implemented in incremental stages, allowing overall system to change and evolve over time

What Is a Requirement?

- Requirement: a statement of what the system must do or what characteristics it needs to have
- Different perspectives:
 - Business requirements – what the business needs
 - User requirements – what users need to do
 - Functional requirements – what the software should do
 - Nonfunctional requirements – characteristics the system should have
 - System requirements – how the system should be built

What Is a Requirement? (cont.)

- During analysis phase, requirements written from perspective of business, focus on what the system needs to do in order to satisfy business user needs
- Starting point: concentrate on what user actually needs to accomplish with system in order to fulfill needed job or task
- By understanding what user needs to do to complete a task, analyst can then determine ways in which new system can support users' needs

What Is a Requirement? (cont.)

- **Functional requirements** relate directly to a process the system should perform as a part of supporting a user task and/or information it should provide as the user is performing a task

What Is a Requirement? (cont.)

- Example
 - User: “The system must retain customer order history for 3 years”
 - Analyst: probe for the reasoning behind
 - Functional requirement: “The system should allow registered customers to review their own order history for the past 3 years” (a process need)

What Is a Requirement? (cont.)

- Example
 - User: “The system should check incoming customer orders for inventory availability” (a process need)
 - Functional requirement: “The system should maintain real-time inventory levels at all warehouses” (information need)

What Is a Requirement? (cont.)

Functional Requirement	Description	Examples
Process-oriented	A process the system must perform; a process the system must do	• The system must allow registered customers to review their own order history for the past 3 years. • The system must check incoming customer orders for inventory availability. • The system should allow students to view a course schedule while registering for classes.
Information-oriented	Information the system must contain	• The system must retain customer order history for 3 years. • The system must include real-time inventory levels at all warehouses. • The system must include budgeted and actual sales and expense amounts for the current year and 3 previous years.

What Is a Requirement? (cont.)

- **Nonfunctional requirements:** the quality attributes, design and implementation constraints, and external interfaces which a product must have

What Is a Requirement? (cont.)

Nonfunctional Requirement	Description	Examples
Operational	The physical and technical environments in which the system will operate	• The system will run on Andriod mobile devices. • The system should be able to integrate with the existing inventory system. • The system should be compatible with any Web browser.
Performance	The speed, capacity, and reliability of the system	• Any interaction between the user and the system should not exceed 2 seconds. • The system downloads new status parameters within 5 minutes of a change. • The system should be available for use 24 hours per day, 365 days per year. • The system supports 300 simultaneous users from 9–11 a.m.; 150 simultaneous users at all other times.

What Is a Requirement? (cont.)

Nonfunctional Requirement	Description	Examples
Security	Who has authorized access to the system under what circumstances	• Only direct managers can see staff personnel records. • Technicians can see only their own work assignments. • The system includes all available safeguards from viruses, worms, Trojan horses, etc.
Cultural and Political	Cultural and political factors and legal requirements that affect the system	• The system should be able to distinguish between US currency and currency from other nations. • Company policy is to buy computers only from Dell. • Country managers are permitted to authorize custom user interfaces within their units. • Personal information is protected in compliance with the Data Protection Act.

What Is a Requirement? (cont.)

CONCEPTS IN ACTION 3-A What Can Happen If You Ignore Nonfunctional Requirements

I once worked on a consulting project in which my manager created a requirements definition without listing nonfunctional requirements. The project was then estimated based on the requirements definition and sold to the client for \$5,000. In my manager's mind, the system that we would build for the client would be a very simple standalone system running on current technology. It should not take more than a week to analyze, design, and build.

Unfortunately, the client had other ideas. They wanted the system to be used by many people in three different departments, and they wanted the ability for any number of people to work on the system concurrently. The technology they had in place was antiquated, but nonetheless they wanted the system to run effectively on the existing equipment. Because we did not set the project scope properly by including our assumptions about nonfunctional requirements in the requirements definition, we basically had to do whatever they wanted.

The capabilities they wanted took weeks to design and program. The project ended up taking 4 months, and the final **project cost** was \$250,000. Our company had to pick up the tab for everything except the agreed upon \$5,000. This was by far the most frustrating project situation I ever experienced. *Barbara Wixom*

Process of Determining Requirements

- Identify primary sources of requirements, including project sponsor, project champion(s), all users of system (both direct and indirect), and possibly others
- Consider how best to elicit requirements from stakeholders
- Analyse information and craft requirements definition statement
- Work with project team and business users to verify, change, and complete list of requirements

Process of Determining Requirements (cont.)

- Prioritise the importance of requirements identified
- Use cases, process models, and data models may be used to clarify and define ideas for new system
- Evolution of requirements definition must be carefully managed

Requirements Definition Statement

- Requirements definition statement (requirements definition) – a straightforward text report that simply lists functional and nonfunctional requirements in outline format

Requirements Definition Statement (cont.)

Functional Requirements

1. Drone Sales Management

- 1.1 The system will enable drone sales order creation.
- 1.2 The system will determine if the requested drone model is in stock.
- 1.3 The system will display all available customization options for a specific drone.
- 1.4 The system will create a final approved sales order.
- 1.5 The system will prepare a shop work order based on final approved configuration.
- 1.6 The system will process a customer deposit.
- 1.7 The system will process a customer final payment.

2. Drone Customization Shop Management

- 2.1 The system will send a Parts Request for needed drone components on an order to Drone Inventory department.
- 2.2 The system enables assignment of a work order to a specific technician.
- 2.3 The system records the arrival of component parts as they arrive in the shop parts room.
- 2.4 The system notifies the assigned technician when all required components are available for a shop work order.
- 2.5 The system enables the technician to record work start time on a work order.
- 2.6 The system allows the technician to record when shop order is completed.
- 2.7 The system notifies the customer of the order completion.

Requirements Definition Statement (cont.)

Nonfunctional Requirements

1. Operational

- 1.1 The system should run on tablet devices to be used by salespeople.
- 1.2 The system should be Web-based and run on any browser.
- 1.3 The system should connect to printers wirelessly.

2. Performance

- 2.1 The system should provide response times of 3 seconds or less.
- 2.2 The system should be updated with new customer orders and drone inventory levels every 5 minutes.

3. Security

- 3.1 Customer accounts should be maintained securely.
- 3.2 Only the customization shop supervisor may approve non-standard customizing options.
- 3.3 Use of each tablet device should be restricted to the salesperson to whom it is assigned.

4. Cultural and Political

- 4.1 Company policy says that all computer equipment is purchased from Dell.

Requirements Definition Statement (cont.)

- Assigning unique numbers enables each requirement to be tracked
- Requirements may be prioritized on the requirements definition statement – “high”, “medium”, “low” importance in new system, or labelled with the version of the system that will address the requirement

Requirements Definition Statement (cont.)

- Purpose of requirements definition:
 - Provide clear statement of what new system should do in order to achieve the system vision described in system request
 - Use cases, process models, and data models provide additional explanatory content
 - Define scope of the system
 - Establish users' expectations for the system
 - Serve as a resource for clarification, in case of discrepancies or misunderstanding